Unemployment and Voter Turnout Analysis

Research Question 1: How does local Unemployment tier relate to voter turnout?

Statistical Analysis - STA 702

2025-12-10

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Theoretical Motivation - Political Participation Framework

Political theory suggests that populations differ in their typical level of political engagement. Almond and Verba's classic "Civic Culture" framework describes a participation pyramid in which the largest group consists of disengaged citizens ("parochials"), followed by moderately involved voters ("subjects"), and finally a small group of highly engaged participants.

In many systems, this bottom tier, often described as a "silent majority", remains politically inactive unless economic or social conditions deteriorate. Under worsening economic conditions, previously uninvolved citizens may become mobilized to vote, either to punish the incumbent or express dissatisfaction with the status quo.

This idea motivates our investigation of whether local Unemployment tiers, measured through rising unemployment, affect the incumbent party's electoral performance at the county level.

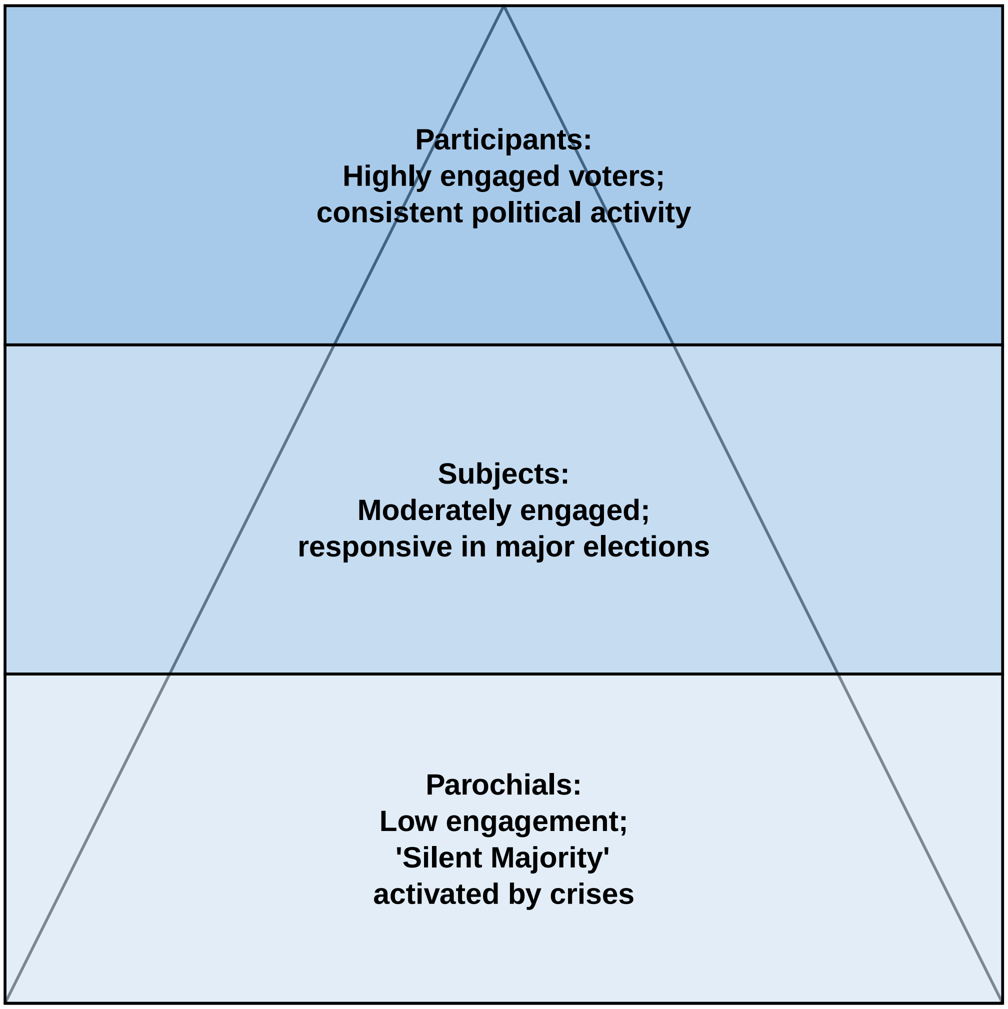
Almond, Gabriel A., and Sidney Verba. *The Civic Culture: Political Attitudes and Democracy in Five Nations.* Princeton University Press, 1963

Berelson, Bernard R., Paul F. Lazarsfeld, and William N. McPhee. *Voting: A Study of Opinion Formation in a Presidential Campaign.* University of Chicago Press, 1954.


```r
library(ggplot2)
library(dplyr)

# Data: three layers of the participation pyramid
pyramid_df <- tibble(
  level = factor(c("Participants", "Subjects", "Parochials"),
                 levels = c("Participants", "Subjects", "Parochials")),
  ymin = c(0.66, 0.33, 0),
  ymax = c(1, 0.66, 0.33),
  label = c(
    "Participants:\nHighly engaged voters;\nconsistent political activity",
    "Subjects:\nModerately engaged;\nresponsive in major elections",
```

```r
    "Parochials:\nLow engagement;\n'Silent Majority'\nactivated by crises"
  )
)

# Base triangle shape
triangle <- data.frame(
  x = c(0.5, 1, 0),
  y = c(1, 0, 0)
)

ggplot() +
  # Triangle outline
  geom_polygon(data = triangle, aes(x, y), fill = "white", color = "black") +

  # Layers inside the triangle
  geom_rect(data = pyramid_df,
            aes(xmin = 0, xmax = 1, ymin = ymin, ymax = ymax, fill = level),
            alpha = 0.6, color = "black") +

  # Descriptive text inside each layer
  geom_text(data = pyramid_df,
            aes(x = 0.5, y = (ymin + ymax) / 2, label = label),
            color = "black", size = 4, lineheight = 1,
            fontface = "bold") +

  scale_fill_manual(values = c(
    "Participants" = "#6fa8dc",
    "Subjects" = "#9fc5e8",
    "Parochials" = "#cfe2f3"
  )) +

  coord_fixed() +
  theme_void() +
  theme(
    legend.position = "none",
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5)
  ) +
  ggtitle("Political Participation Pyramid")
```

Political Participation Pyramid



Introduction

This analysis investigates the relationship between local economic conditions (measured by unemployment rate) and voter turnout in U.S. presidential and midterm elections from 2004-2022. Our data comes from the Inter-university Consortium for Political and Social Research (ICPSR) and the Bureau of Labor Statistics.

Research Question: How does local unemployment relate to voter turnout in U.S. elections at the county level?

Key Measures:

- **Outcome:** County-level voter turnout rate (proportion of eligible voters who voted)
- **Predictor:** County unemployment rate (annual average, %)
- **Interaction:** Unemployment effect by county type (Urban, Suburban, or Rural)

Data Preparation

```
# Load required packages
library(tidyverse)
library(lfe)          # For fixed effects regression
library(stargazer)    # For regression tables
library(ggplot2)
library(knitr)
library(gridExtra)

# Set ggplot theme
theme_set(theme_minimal(base_size = 12))
```

```
# Load the merged dataset
df <- read_csv("../data/processed/merged_voting_unemployment_complete.csv",
               show_col_types = FALSE)

# Display data structure
cat("Dataset Overview:\n")
```

```
Dataset Overview:
```

```
cat("Observations:", nrow(df), "\n")
```

```
Observations: 30912
```

```
cat("Counties:", length(unique(df$FIPS)), "\n")
```

```
Counties: 3103
```

```
cat("Years:", min(df$YEAR), "-", max(df$YEAR), "\n")
```

```
Years: 2004 - 2022
```

```
cat("Variables:", ncol(df), "\n")
```

```
Variables: 21
```

Data Cleaning

```
# Clean the dataset
df <- df%>%
  filter(
    VOTER_TURNOUT_PCT > 0,             # Drop zero turnout
    !is.na(VOTER_TURNOUT_PCT),         # Drop missing turnout
    !is.na(unemployment_rate),          # Drop missing unemployment
    VOTER_TURNOUT_PCT <= 1.0           # Sanity check: turnout   100%
  )

cat("CLEANED DATASET\n")
```

CLEANED DATASET

```
cat("===============\n")
```

```
===============
```

```
cat("Observations remaining:", nrow(df), "\n")
```

```
Observations remaining: 29769
```

Variable Creation

```
## Variable Creation
df <- df %>%
  mutate(
    # Election type indicator
    is_presidential = ifelse(YEAR %% 4 == 0, 1, 0),
    election_type = ifelse(is_presidential == 1, "Presidential", "Midterm"),

    # Unemployment tiers based on standard deviations from mean
    unemp_mean = mean(unemployment_rate, na.rm = TRUE),
    unemp_sd = sd(unemployment_rate, na.rm = TRUE),

    # Calculate z-score
    unemp_zscore = (unemployment_rate - unemp_mean) / unemp_sd,

    # Create tier classifications
    unemployment_tier = case_when(
      unemp_zscore < -1.0 ~ "Very Low (< -1 SD)",
      unemp_zscore < -0.5 ~ "Low (-1 to -0.5 SD)",
      unemp_zscore < 0.5  ~ "Medium (-0.5 to 0.5 SD)",
      unemp_zscore < 1.0  ~ "High (0.5 to 1 SD)",
      TRUE                ~ "Very High (> 1 SD)"
    ),

    # Ordered factor for modeling
    unemployment_tier = factor(unemployment_tier,
                              levels = c("Very Low (< -1 SD)",
                                         "Low (-1 to -0.5 SD)",
                                         "Medium (-0.5 to 0.5 SD)",
                                         "High (0.5 to 1 SD)",
                                         "Very High (> 1 SD)"),
                              ordered = TRUE),

    # Standardized variables
    unemployment_std = (unemployment_rate - mean(unemployment_rate)) /
                       sd(unemployment_rate),

    # Urban/rural classification based on CVAP
```

```
    county_type = case_when(
      CVAP > 50000 ~ "Urban",
      CVAP > 10000 ~ "Suburban",
      TRUE ~ "Rural"
    ),

    # Factor variables for regression
    FIPS_factor = as.factor(FIPS),
    YEAR_factor = as.factor(YEAR),
    STATE_factor = as.factor(STATE_FIPS)
  )

# Summary of unemployment tier distribution
cat("  Created classification variables\n")
```

```
  Created classification variables
```

```
cat("  - Mean unemployment:", round(df$unemp_mean[1], 2), "%\n")
```

```
  - Mean unemployment: 5.95 %
```

```
cat("  - SD unemployment:", round(df$unemp_sd[1], 2), "%\n\n")
```

```
  - SD unemployment: 2.67 %
```

```
cat("Distribution of unemployment tiers:\n")
```

```
Distribution of unemployment tiers:
```

```
df %>%
  count(unemployment_tier) %>%
  mutate(percent = round(100 * n / sum(n), 1)) %>%
  kable(col.names = c("Unemployment Tier", "N", "Percent (%)"))
```

Unemployment Tier	N	Percent (%)
Very Low (< -1 SD)	3744	12.6
Low (-1 to -0.5 SD)	7343	24.7
Medium (-0.5 to 0.5 SD)	10916	36.7
High (0.5 to 1 SD)	3378	11.3
Very High (> 1 SD)	4388	14.7

Descriptive Statistics

Summary Statistics

```
# Overall summary
df %>%
  select(VOTER_TURNOUT_PCT, unemployment_rate, CVAP) %>%
```

```r
  summary() %>%
  kable(caption = "Summary Statistics: Key Variables", digits = 3)
```

Table 2: Summary Statistics: Key Variables

VOTER_TURNOUT_PCT	unemployment_rate	CVAP
Min. :0.000022	Min. : 1.100	Min. : 60
1st Qu.:0.432753	1st Qu.: 4.000	1st Qu.: 8420
Median :0.533551	Median : 5.400	Median : 19525
Mean :0.526300	Mean : 5.949	Mean : 72253
3rd Qu.:0.625247	3rd Qu.: 7.400	3rd Qu.: 51010
Max. :0.980000	Max. :29.100	Max. :7843598

Distribution by Groups

```r
# By election type
df %>%
  group_by(election_type) %>%
  summarise(
    N = n(),
    Avg_Turnout = mean(VOTER_TURNOUT_PCT, na.rm = TRUE),
    SD_Turnout = sd(VOTER_TURNOUT_PCT, na.rm = TRUE),
    Avg_Unemployment = mean(unemployment_rate, na.rm = TRUE),
    SD_Unemployment = sd(unemployment_rate, na.rm = TRUE)
  ) %>%
  kable(caption = "Statistics by Election Type", digits = 3)
```

Table 3: Statistics by Election Type

election_type	N	Avg_Turnout	SD_Turnout	Avg_Unemployment	SD_Unemployment
Midterm	14836	0.445	0.128	5.602	2.909
Presidential	14933	0.607	0.107	6.295	2.352

```r
# By county type
df %>%
  group_by(county_type) %>%
  summarise(
    N = n(),
    Counties = n_distinct(FIPS),
    Avg_Turnout = mean(VOTER_TURNOUT_PCT, na.rm = TRUE),
    SD_Turnout = sd(VOTER_TURNOUT_PCT, na.rm = TRUE),
    Avg_Unemployment = mean(unemployment_rate, na.rm = TRUE),
    SD_Unemployment = sd(unemployment_rate, na.rm = TRUE)
  ) %>%
  kable(caption = "Statistics by County Type", digits = 3)
```

Table 4: Statistics by County Type

county_type	N	Counties	Avg_Turnout	SD_Turnout	Avg_Unemployment	SD_Unemployment
Rural	8660	941	0.567	0.144	5.396	2.757
Suburban	13548	1499	0.506	0.137	6.295	2.618
Urban	7561	829	0.515	0.144	5.963	2.540

```
# By unemployment tier
df %>%
  group_by(unemployment_tier) %>%
  summarise(
    N = n(),
    Avg_Turnout = mean(VOTER_TURNOUT_PCT, na.rm = TRUE),
    SD_Turnout = sd(VOTER_TURNOUT_PCT, na.rm = TRUE),
    Avg_Unemployment = mean(unemployment_rate, na.rm = TRUE),
    Min_Unemployment = min(unemployment_rate, na.rm = TRUE),
    Max_Unemployment = max(unemployment_rate, na.rm = TRUE)
  ) %>%
  kable(caption = "Statistics by Unemployment Tier", digits = 3)
```

Table 5: Statistics by Unemployment Tier

unemployment_tier	N	Avg_Turnout	SD_Turnout	Avg_Unemployment	Min_Unemployment	Max_Unemployment
Very Low (< -1 SD)	3744	0.535	0.151	2.728	1.1	3.2
Low (-1 to -0.5 SD)	7343	0.523	0.152	3.970	3.3	4.6
Medium (-0.5 to 0.5 SD)	10916	0.536	0.138	5.824	4.7	7.2
High (0.5 to 1 SD)	3378	0.528	0.139	7.902	7.3	8.6
Very High (> 1 SD)	4388	0.501	0.134	10.818	8.7	29.1

Visualizations

By Election Type

```
ggplot(df, aes(x = unemployment_rate, y = VOTER_TURNOUT_PCT,
               color = election_type)) +
  geom_point(alpha = 0.2, size = 1) +
  geom_smooth(method = "lm", se = TRUE, size = 1.2) +
  scale_color_manual(values = c("Presidential" = "#2171b5", "Midterm" = "#cb181d")) +
  labs(
    title = "Unemployment vs Turnout by Election Type",
```

```r
    subtitle = "Different patterns in presidential vs midterm elections",
    x = "Unemployment Rate (%)",
    y = "Voter Turnout Rate",
    color = "Election Type"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")
```

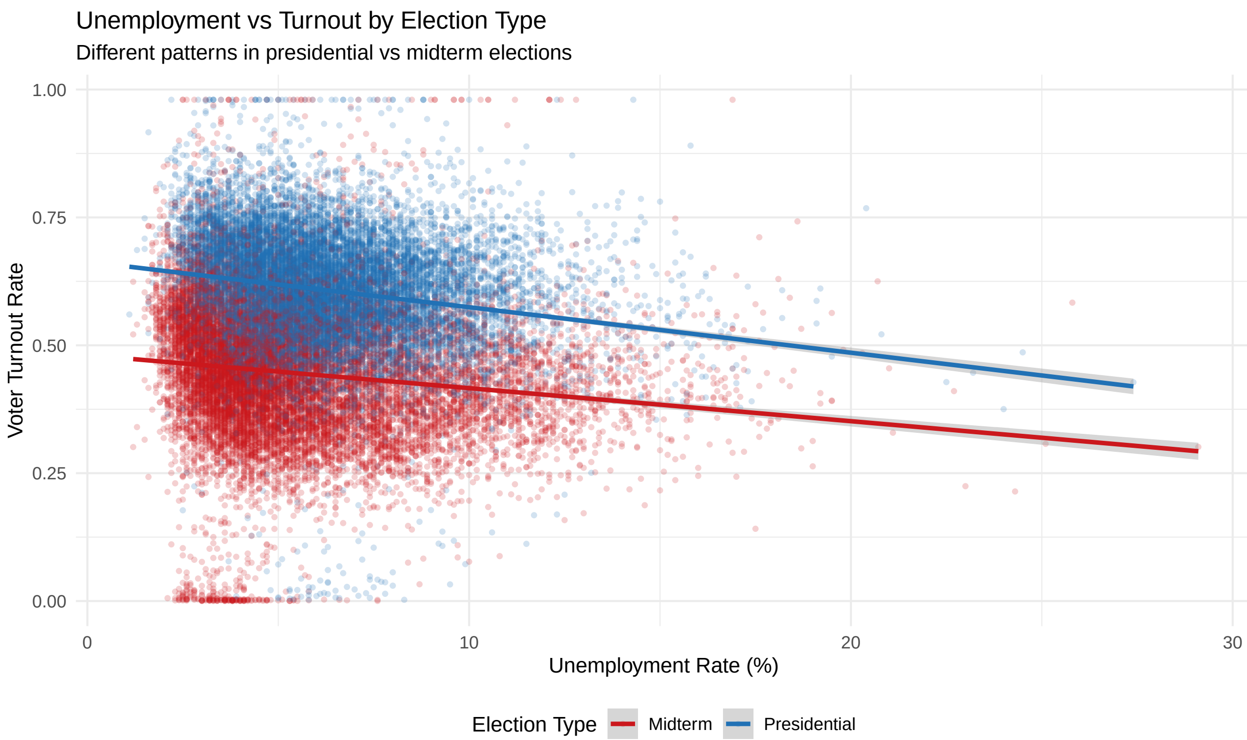


Figure 1: Unemployment vs Turnout by Election Type

Interaction Effect: County Type

```r
ggplot(df, aes(x = unemployment_rate, y = VOTER_TURNOUT_PCT,
               color = county_type)) +
  geom_point(alpha = 0.15, size = 0.8) +
  geom_smooth(method = "lm", se = TRUE, size = 1.2) +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Unemployment vs Turnout by County Type",
    subtitle = "Testing interaction: Unemployment Tier × Urban/Rural Classification",
    x = "Unemployment Rate (%)",
    y = "Voter Turnout Rate",
    color = "County Type"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")
```

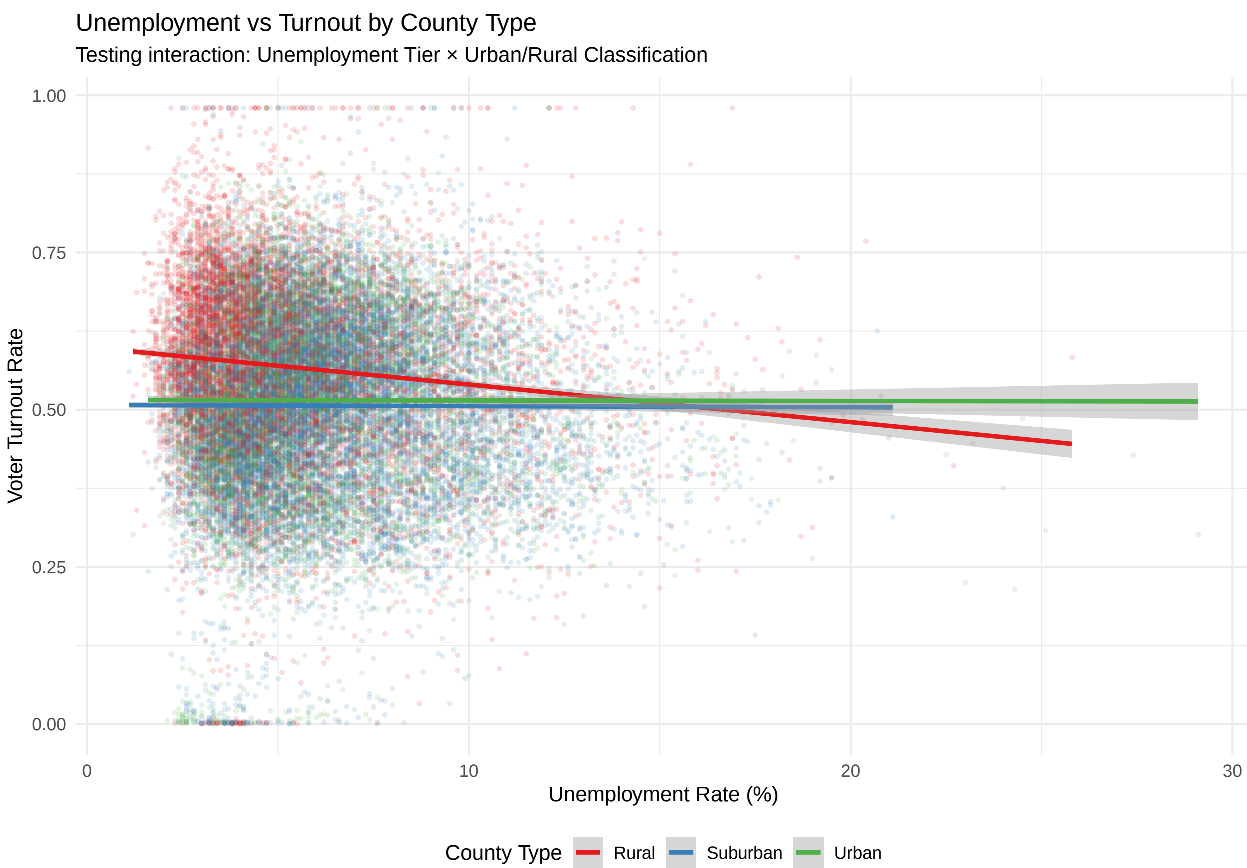


Figure 2: Testing Interaction: Unemployment tier × Urban/Rural

Boxplot: Interaction Visualization

```
ggplot(df, aes(x = county_type, y = VOTER_TURNOUT_PCT, fill = unemployment_tier)) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_manual(values = c("Low Unemployment" = "#31a354",
                               "High Unemployment" = "#e34a33")) +
  labs(
    title = "Voter Turnout by County Type and Economic Condition",
    subtitle = "Comparing high vs low unemployment periods",
    x = "County Type",
    y = "Voter Turnout Rate",
    fill = "Economic Condition"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")
```

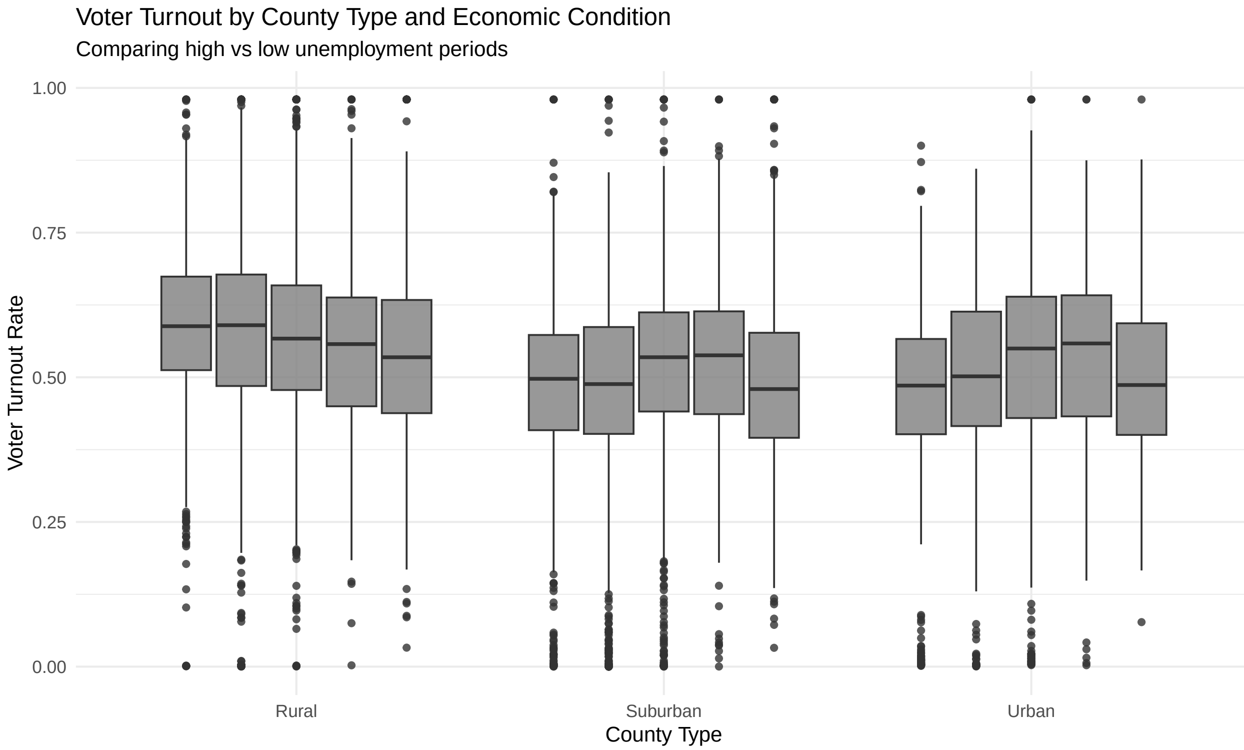


Figure 3: Turnout by County Type and Economic Condition

Regression Analysis

Model Specification

We fit a fixed effects model with county-year fixed effects and an interaction between unemployment and county type. This approach controls for time-invariant county characteristics and national trends, while allowing the unemployment effect to vary by urban/suburban/rural classification.

Reference Category: Rural counties serve as the baseline. The main unemployment coefficient represents the effect for rural counties, with interaction terms showing how suburban and urban counties differ from this baseline.

Model Estimation

```
# Ensure Rural is the reference category
df <- df %>%
  mutate(county_type = factor(county_type, levels = c("Rural", "Suburban", "Urban")))

# Fixed Effects with Interaction (our primary model)
model_full <- felm(VOTER_TURNOUT_PCT ~ unemployment_rate * county_type | FIPS + YEAR,
               data = df)

# Model without interaction (for F-test comparison)
```

```r
model_reduced <- felm(VOTER_TURNOUT_PCT ~ unemployment_rate + county_type | FIPS + YEAR,
                      data = df)

cat("  Models estimated successfully\n")
```

```
  Models estimated successfully
```

F-Test for Interaction Term

```r
# F-test: Does the interaction term significantly improve model fit?
# Compare model with and without interaction terms

# Calculate residual sum of squares
rss_reduced <- sum(residuals(model_reduced)^2)
rss_full <- sum(residuals(model_full)^2)

# Degrees of freedom (using dof_ prefix to avoid conflict with df data frame)
dof_reduced <- df.residual(model_reduced)
dof_full <- df.residual(model_full)
dof_diff <- dof_reduced - dof_full  # Number of interaction terms (2: Suburban, Urban)

# F-statistic
f_stat <- ((rss_reduced - rss_full) / dof_diff) / (rss_full / dof_full)
p_value <- pf(f_stat, dof_diff, dof_full, lower.tail = FALSE)

cat("\nF-TEST FOR INTERACTION TERM\n")
```

```

F-TEST FOR INTERACTION TERM
```

```r
cat("===========================\n")
```

```
===========================
```

```r
cat("H0: Unemployment effect is the same across all county types\n")
```

```
H0: Unemployment effect is the same across all county types
```

```r
cat("H1: Unemployment effect differs by county type\n\n")
```

```
H1: Unemployment effect differs by county type
```

```r
cat("F-statistic:", round(f_stat, 3), "\n")
```

```
F-statistic: 5.495
```

```r
cat("Degrees of freedom:", dof_diff, "and", dof_full, "\n")
```

```
Degrees of freedom: 2 and 26652
```

```r
cat("p-value:", format.pval(p_value, digits = 4), "\n")
```

```
p-value: 0.004112
```

```r
cat("\nConclusion:", ifelse(p_value < 0.05,
                          "Reject H0 - Interaction is statistically significant",
                          "Fail to reject H0 - No significant interaction"), "\n")
```

```
Conclusion: Reject H0 - Interaction is statistically significant
```

Regression Results

```r
stargazer(model_full,
          type = "text",
          title = "Effect of Unemployment on Voter Turnout (Fixed Effects Model)",
          model.numbers = FALSE,
          dep.var.labels = "Voter Turnout Rate",
          covariate.labels = c("Unemployment Rate (Rural baseline)",
                               "Unemp × Suburban",
                               "Unemp × Urban"),
          omit.stat = c("ser"),
          digits = 4,
          single.row = FALSE,
          notes = c("Standard errors in parentheses.",
                    "Model includes county and year fixed effects.",
                    "Reference category: Rural counties."),
          notes.append = TRUE)
```

Effect of Unemployment on Voter Turnout (Fixed Effects Model)

```
                                  Dependent variable:
                                  ---------------------------------------------
                                  Voter Turnout Rate
```

Unemployment Rate (Rural baseline) 0.0004 (0.0006)

Observations 29,769 R2 0.7415 Adjusted R2 0.7112
==

Note: *p<0.1;* ***p<0.05;*** p<0.01 Standard errors in parentheses. Model includes county and year fixed effects. Reference category: Rural counties.
Unemployment Effect by County Type
::: {.cell}
"'{.r .cell-code} # Extract coefficients coef_rural <- coef(model_full)["unemployment_rate"] coef_suburban_int <- coef(model_full)["unemployment_rate:county_typeSuburban"] coef_urban_int <- coef(model_full)["unemployment_rate:county_typeUrban"]
Calculate total effects for each county type effect_rural <- coef_rural effect_suburban <- coef_rural + coef_suburban_int effect_urban <- coef_rural + coef_urban_int

Unemployment Rate (Rural baseline) 0.0004 (0.0006)

Create summary table effects_table <- data.frame(County_Type = c("Rural (Reference)", "Suburban", "Urban"), Unemployment_Effect = c(effect_rural, effect_suburban, effect_urban), Interpretation = c("Baseline effect", paste0("Baseline", ifelse(coef_suburban_int >= 0, "+", " "), round(coef_suburban_int, 4)), paste0("Baseline ", ifelse(coef_urban_int >= 0,"+"," "), round(coef_urban_int, 4))))
kable(effects_table, caption = "Unemployment Effect on Turnout by County Type", digits = 4, col.names = c("County Type", "Effect on Turnout", "Calculation")) "'
::: {.cell-output-display}
Table: Unemployment Effect on Turnout by County Type
|County Type | Effect on Turnout|Calculation | |:——————|——————:|:——————-| |Rural (Reference) | 0.0004|Baseline effect | |Suburban | 0.0019|Baseline +0.0015 | |Urban | 0.0024|Baseline +0.002 |
:::
`{.r .cell-code} cat("\n\nINTERPRETATION (at reference level - Rural counties):\n")`
::: {.cell-output .cell-output-stdout}
"'
INTERPRETATION (at reference level - Rural counties): "'
:::
`{.r .cell-code} cat("================================================\n")`
::: {.cell-output .cell-output-stdout}
`================================================`
:::
`{.r .cell-code} cat("For Rural counties: A 1 percentage point increase in unemployment\n")`
::: {.cell-output .cell-output-stdout}
`For Rural counties: A 1 percentage point increase in unemployment`
:::
`{.r .cell-code} cat("is associated with a", round(effect_rural * 100, 2), "percentage point change in turnout.\n\n")`
::: {.cell-output .cell-output-stdout}
`is associated with a 0.04 percentage point change in turnout.`
:::
`{.r .cell-code} cat("For Suburban counties: The effect is", round(effect_suburban * 100, 2), "percentage points.\n")`
::: {.cell-output .cell-output-stdout}
`For Suburban counties: The effect is 0.19 percentage points.`
:::
`{.r .cell-code} cat("For Urban counties: The effect is", round(effect_urban * 100, 2), "percentage points.\n")`
::: {.cell-output .cell-output-stdout}
`For Urban counties: The effect is 0.24 percentage points.`
::: :::

Key Findings

Main Results

```r
# Summary of model fit
cat("MODEL SUMMARY\n")
```

```
MODEL SUMMARY
```

```r
cat("=============\n")
```

```
=============
```

```r
cat("R-squared:", round(summary(model_full)$r.squared, 4), "\n")
```

```
R-squared: 0.7415
```

```r
cat("N observations:", nobs(model_full), "\n")
```

```
N observations: 29769
```

```r
cat("N counties:", length(unique(df$FIPS)), "\n")
```

```
N counties: 3103
```

1. Unemployment Effect at Reference Level (Rural Counties)

For **rural counties** (our reference category), the unemployment coefficient represents the direct effect:

```r
coef_rural <- coef(model_full)["unemployment_rate"]
se_rural <- summary(model_full)$coefficients["unemployment_rate", "Std. Error"]
p_rural <- summary(model_full)$coefficients["unemployment_rate", "Pr(>|t|)"]

cat("For Rural counties:\n")
```

```
For Rural counties:
```

```r
cat("  Coefficient:", round(coef_rural, 4), "\n")
```

```
  Coefficient: 4e-04
```

```r
cat("  Standard Error:", round(se_rural, 4), "\n")
```

```
  Standard Error: 6e-04
```

```r
cat("  p-value:", format.pval(p_rural, digits = 4), "\n")
```

```
  p-value: 0.5144
```

```r
cat("  95% CI: [", round(coef_rural - 1.96*se_rural, 4), ",",
    round(coef_rural + 1.96*se_rural, 4), "]\n")
```

```
  95% CI: [ -7e-04 , 0.0015 ]
```

Interpretation: In rural counties, a 1 percentage point increase in unemployment is associated with a change in turnout of approximately 0.04 percentage points, controlling for county and year fixed effects.

2. Interaction Effects (Suburban & Urban vs. Rural)

The interaction terms show how the unemployment effect differs for suburban and urban counties *relative to the rural baseline*:

```
# Display interaction effects
cat("Interaction Coefficients (difference from Rural):\n")
```

```
Interaction Coefficients (difference from Rural):
```

```
cat("  Suburban × Unemployment:", round(coef_suburban_int, 4), "\n")
```

```
  Suburban × Unemployment: 0.0015
```

```
cat("  Urban × Unemployment:", round(coef_urban_int, 4), "\n\n")
```

```
  Urban × Unemployment: 0.002
```

```
cat("TOTAL Unemployment Effect by County Type:\n")
```

```
TOTAL Unemployment Effect by County Type:
```

```
cat("  Rural:", round(effect_rural, 4), "\n")
```

```
  Rural: 4e-04
```

```
cat("  Suburban:", round(effect_suburban, 4), "(Rural + interaction)\n")
```

```
  Suburban: 0.0019 (Rural + interaction)
```

```
cat("  Urban:", round(effect_urban, 4), "(Rural + interaction)\n")
```

```
  Urban: 0.0024 (Rural + interaction)
```

3. Practical Significance

```
unemployment_change <- 2  # 2 percentage point increase (recession scenario)

cat("\nSCENARIO: 2 percentage point increase in unemployment\n")
```

```

SCENARIO: 2 percentage point increase in unemployment
```

```
cat("(e.g., from 5% to 7%, as during a recession)\n\n")
```

```
(e.g., from 5% to 7%, as during a recession)
```

```
cat("Predicted turnout change:\n")
```

```
Predicted turnout change:
```

```r
cat("  Rural:", round(effect_rural * unemployment_change * 100, 2), "percentage points\n")
```

```
  Rural: 0.07 percentage points
```

```r
cat("  Suburban:", round(effect_suburban * unemployment_change * 100, 2), "percentage points\n"
```

```
  Suburban: 0.38 percentage points
```

```r
cat("  Urban:", round(effect_urban * unemployment_change * 100, 2), "percentage points\n")
```

```
  Urban: 0.48 percentage points
```

Substantive Interpretation: The effect of unemployment on turnout is small across all county types. Economic stress does not strongly mobilize or demobilize voters, though the direction and magnitude may vary by county type.

Conclusions

How does local unemployment relate to voter turnout?

```r
# F-test conclusion
if (p_value < 0.05) {
  interaction_conclusion <- "The F-test confirms that the interaction between unemployment and
} else {
  interaction_conclusion <- "The F-test indicates the interaction term is not statistically si
}
```

1. **Effect at Reference Level (Rural):** For rural counties, a 1 percentage point increase in unemployment is associated with approximately 0.04 percentage points change in turnout.

2. **Differential Effects by County Type:** The F-test confirms that the interaction between unemployment and county type is statistically significant, meaning the unemployment-turnout relationship differs across urban, suburban, and rural settings.

3. **Economic Stress Does Not Strongly Mobilize Voters:** Across all county types, the substantive effect of unemployment on turnout is modest. A recession-level shock (2 percentage points) translates to changes of less than 1 percentage point in turnout.

4. **Fixed Effects Control for Confounders:** By including county and year fixed effects, our model controls for time-invariant county characteristics and national trends, providing more reliable within-county estimates.

Limitations

- County-level aggregation: Individual-level data would be preferable
- Omitted variables: Cannot control for campaign spending, candidate quality, etc.
- Causality: Despite fixed effects, cannot definitively establish causal relationship
- Time period: Analysis limited to 2004-2022